ANALYSIS REPORT

Polished Early-Wave Parallel Training Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `polished_dataset_early_wave_parallel_training_2026_06_25` campaign completed the approved early-wave parallel retraining batch with `36` completed runs and `0` failed runs.

The accepted campaign output directory is `output/training_campaigns/2026-06-25-16-01-23_polished_dataset_early_wave_parallel_training_2026_06_25`.

The scalar leaderboard winner is `te_periodic_gru_sequence_bw`, trained on `polished_dataset` with the `polished_point_v1` schema:

- surface: `bw` / `backward_only`;
- test MAE: `0.001084 deg`;
- test RMSE: `0.001393 deg`;
- validation MAE: `0.001088 deg`;
- trainable parameters: `157,569`.

  This is a normal campaign closeout. It accepts the completed model-development training batch and synchronizes campaign status. It does not promote a new official compensation candidate through the `TE Curve Verification Pipeline`; that curve-first verification remains a separate operator-approved workflow.

Campaign Scope

The campaign reused the first 36 prepared configurations from the larger `polished_dataset_full_wave_retraining_2026_06_22` package. It covered `12` model families across `global`, `fw`, and `bw` surfaces.

All accepted runs used the polished input columns:

Field	Role
`theta`	motor position measured in degrees
`theta_dot`	motor velocity derived from position
`tau_load`	applied load in Nm
`T`	oil temperature
`theta_TE`	measured transmission error target

Included model families

- `feedforward`
- `gru_sequence`
- `harmonic_regression`
- `lstm_sequence`
- `periodic_gru_sequence`

- `periodic_lstm_sequence`
- `periodic_mlp`
- `periodic_mlp_harmonic`
- `periodic_temporal_convolution`
- `residual_harmonic_mlp`
- `temporal_convolution`
- `tree`

Queue And Artifact Audit

Check	Result
Completed queue entries	`36`
Failed queue entries	`0`
Running queue entries	`0`
Pending queue entries	`0`
Leaderboard entries	`36`
Dataset entries recorded as `polished_dataset`	`36`
Schema entries recorded as `polished_point_v1`	`36`
Missing referenced metrics/reports/checkpoints	`0`

The run logs contain no campaign-level failures. The repeated `Error Message: N/A` lines in the execution report belong to completed run-detail records, not to actual failures.

Surface Winners

Surface	Best Run	Family	Test MAE	Test RMSE	Val MAE	Params
Global	`te_periodic_lstm_sequence_global`	`periodic_lstm_sequence`	0.001187	0.001505	0.001185	210,049
Forward-only	`te_periodic_gru_sequence_fw`	`periodic_gru_sequence`	0.001101	0.001409	0.001099	157,569
Backward-only	`te_periodic_gru_sequence_bw`	`periodic_gru_sequence`	0.001084	0.001393	0.001088	157,569

Scalar Leaderboard Snapshot

The table below shows the first eight entries from the scalar leaderboard. The full ordered list, with exact run names and artifact paths, is stored in `output/training_campaigns/2026-06-25-16-01-23_polished_dataset_early_wave_parallel_training_2026_06_25/campaign_leaderboard.yaml` .

Rank	Compact Run	Surface	Family	Test MAE	Test RMSE	Val MAE
1	`periodic_gru_bw`	`bw`	`periodic_gru`	0.001084	0.001393	0.001088
2	`periodic_gru_fw`	`fw`	`periodic_gru`	0.001101	0.001409	0.001099
3	`periodic_lstm_global`	`global`	`periodic_lstm`	0.001187	0.001505	0.001185
4	`periodic_gru_global`	`global`	`periodic_gru`	0.001257	0.001613	0.001252
5	`periodic_mlp_harmonic_global`	`global`	`periodic_mlp_harmonic`	0.001264	0.001737	0.001196
6	`periodic_mlp_harmonic_bw`	`bw`	`periodic_mlp_harmonic`	0.001279	0.001719	0.001103
7	`periodic_mlp_harmonic_fw`	`fw`	`periodic_mlp_harmonic`	0.001326	0.001780	0.001144
8	`periodic_lstm_bw`	`bw`	`periodic_lstm`	0.001338	0.001719	0.001231

Registry And Program Effects

The completed run updated family-level registries for the early-wave polished families and the program scalar registry. The current scalar program winner is `te_periodic_gru_sequence_bw` by campaign `test_mae` .

This scalar winner is useful as a retraining result, but it does not replace the official direction-parallel curve-verified leaders until a separate `TE Curve Verification Pipeline` refresh is run and accepted. The repository must keep `global` , `Fw` , and `Bw` surfaces visible as separate decision surfaces.

Acceptance Decision

The early-wave campaign is accepted as a completed polished-dataset retraining batch because:

- all `36` selected configs completed;
- no queue entries remain pending, running, or failed;
- every leaderboard entry records `polished_dataset` ;
- every leaderboard entry records `polished_point_v1` ;
- every accepted model uses four input features and one TE target;
- every referenced metric, training report, and checkpoint artifact exists.

Closeout Notes

The active campaign state has to preserve the parallel `polished_dataset_rcim_model_bank_reproduction_2026_06_22` provenance because that RCIM campaign was launched on another workstation and is tracked separately from this local early-wave batch.

The TE Program Status And Closeout Ledger was checked. It requires a content update because the latest normal campaign closeout and scalar program winner changed from the earlier Stage 1 smoke result to this completed early-wave parallel training result.

Follow-Up

Recommended next steps

1. finish or inspect the parallel RCIM Model-Bank Reproduction campaign when the other workstation reports completion;
2. keep the remaining 72-run polished full-wave retraining package as the next model-development training stage;
3. after normal campaign closeouts are complete, prepare a separate operator-run `TE Curve Verification Pipeline` refresh for selected polished-trained candidates.